

Test #1 of Erlang

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1. Write a function `rmlst/1` (*remove the last*) such that if

$$\text{rmlst}(P) \xrightarrow{+} Q,$$

then Q is the list L without its last item. For example,

$$\text{rmlst}([1, 2, 3]) \xrightarrow{+} [1, 2].$$

If the list is empty, the result is the atom `empty`: $\text{rmlst}([]) \xrightarrow{+} \text{empty}$.

2. Write a function `pair/2` such that if

$$\text{zip}(P, Q) \xrightarrow{+} R,$$

then R is the list made of the pairs of items of lists P and Q . For example,

$$\text{zip}([1, 2, 3], [\text{a}, \text{b}, \text{c}]) \xrightarrow{+} [\{1, \text{a}\}, \{2, \text{b}\}, \{3, \text{c}\}].$$

If one list is longer than the other, then the atom `length` should be the result, e.g., $\text{zip}([1, 2], [\text{a}]) \xrightarrow{+} \text{length}$.

3. Write a function `twoway/1` such that if the input list is the same when read from right to left, the result is `true`, otherwise it is `false`. For instance,

$$\text{twoway}([0, 1, 0, 1, 0]) \xrightarrow{+} \text{true},$$

$$\text{twoway}([0, 1, 1, 0]) \xrightarrow{+} \text{true},$$

$$\text{twoway}([\text{r}, \text{a}, \text{d}, \text{a}, \text{r}]) \xrightarrow{+} \text{true},$$

$$\text{twoway}([0, 1, 1, 0, 0, 1, 0]) \xrightarrow{+} \text{false},$$

$$\text{twoway}([]) \xrightarrow{+} \text{false}.$$